

CSCE-636: Homework 1

Due on September 17, 2025 at 11:59 pm

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Problem 1

Check out the Jupyter notebook for Chapter 2 at https://github.com/fchollet/deep-learning-with-python-notebooks/blob/master/chapter02_mathematical-building-blocks.ipynb.

Then:

For the neural network in the Jupiter notebook for the MNIST task (which has 2 layers, where the first layer has 512 neurons and the second layer has 10 neurons), adjust the size of the first layer (namely, the number of neurons in the first layer) as: 16, 32, 64, 128, 256, 512. For each size, train the neural network and record its test accuracy (namely, accuracy on the test set). Draw a figure, where the x-axis is the size of the first layer, and the y-axis is the test accuracy. Submit your figure.

Solution

hey

Problem 2

For the neural network in the Jupiter notebook for the MNIST task (which has 2 layers, where the first layer has 512 neurons and the second layer has 10 neurons), adjust the number of layers to be: 2, 3, 4, 5. (The last layer has to have size 10 since we have 10 classes. But you can decide on the sizes of the other layers.) For each case, train the neural network and record its test accuracy. Draw a figure, where the x-axis is the number of layers, and the y-axis is the test accuracy. Submit your figure, and specify the sizes of your layers for each case.

Solution

hey